

Time spent: 0 sec
Subject:
General Chemistry

UID: 401308
System:
Solutions and Electrochemistry

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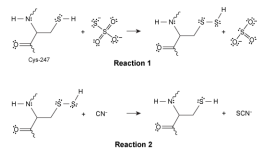
The impact of a given exposure depends on the amount (eg, dose, duration), the chemical composition of the cyanide compound (eg, HCN, NaCN, KCN), and the route of exposure (eg, inhalation, absorption, ingestion). The ionic salts (NaCN, KCN) and free acid (HCN) forms of cyanide each have different acid-base properties that interact differently under the physiological conditions associated with each route of exposure.

To better understand the toxicity of different forms of cyanide, animal models have been studied for comparison to known human exposure case studies. Table 1 presents a summary of selected animal toxicity studies that examine exposure via ingestion of different cyanide compounds. The table provides the lethal dose to 50% of the test group (LD₅₀) in milligrams of cyanide compound per kilogram of body weight (mg/kg). The equivalent molar dose in millimoles per kilogram is also reported.

Table 1 Compilation of LD₅₀ Values for Cyanide Compounds in Various Animal Species

Animal species	LD ₅₀ (mg/kg)	Molar dose (mmol/kg)	CN ⁻ compound
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Based on animal models, one treatment developed for cyanide poisoning involves assisting the rhodanese enzyme in metabolizing CN⁻ to SCN⁻ via an intravenous solution of NaNO₂ and Na₂S₂O₃. The Na₂S₂O₃ assists in the formation of a reactive S–SH (persulfide) bond in the enzyme on cysteine-247 (Reaction 1), which then reacts with the cyanide ion to form thiocyanate (Reaction 2).

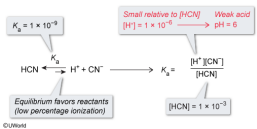


In an aqueous solution of 1×10^{-3} M HCN ($K_a \approx 1 \times 10^{-9}$), HCN can best be described as a:

- ☒ A. weak acid, because $[H^+] < 1 \times 10^{-3}$ M. [50%]
☐ B. strong acid, because $[H^+] = 1 \times 10^{-9}$ M. [26%]
☐ C. weak base, because $[H^+] < 1 \times 10^{-7}$ M. [17%]
☐ D. strong base, because $[H^+] \approx 1 \times 10^{-9}$ M. [5%]

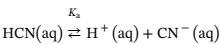
Correct
50% answered correctly

Explanation:



A large value for the **acid dissociation constant** ($K_a > 1$) indicates that a reaction favors the products whereas a small value indicates that a reaction favors the reactants. A small acid dissociation constant ($K_a < 1$) indicates that the ionization equilibrium favors the non-ionized HA reactant. Because **strong acids** ionize essentially 100% in aqueous solutions, they have large **values for the ionization constant** whereas weak acids have small K_a values.

The **small** K_a value for HCN ($K_a \approx 1 \times 10^{-9}$) indicates that it is a **weak acid** and the **non-ionized** form is **favored**. As such, only a small amount (x) of HCN(aq) ionizes.



In a 1×10^{-3} M HCN(aq) solution, the concentrations of each species are summarized as:

	[HCN]	[H ⁺]	[CN ⁻]
Initial:	0.001	0	0
Change:	-x	+x	+x
Equilibrium:	0.001 - x	x	x

Substituting the equilibrium concentrations $[H^+]$, $[CN^-]$, and $[HCN]$ into the K_a equation gives:

$$K_a = \frac{[H^+][CN^-]}{[HCN]}$$
$$1 \times 10^{-9} = \frac{(x)(x)}{(1 \times 10^{-3} - x)} = \frac{x^2}{(0.001 - x)}$$

When x is small, the expression can be simplified by approximation as:

$$1 \times 10^{-9} = \frac{x^2}{(1 \times 10^{-3} - x)} \approx \frac{x^2}{1 \times 10^{-3}}$$

Solving for x gives $[H^+]$:

$$x^2 = (1 \times 10^{-9})(1 \times 10^{-3}) = 1 \times 10^{-12}$$

$$x = \sqrt{1 \times 10^{-12}} = 1 \times 10^{-6} = [H^+]$$

Therefore, $[H^+] = 1 \times 10^{-6}$ M, which is much less than the initial HCN concentration (1×10^{-3} M) and confirms that only a small percentage of the dissolved HCN molecules in solution are ionized (which is consistent with a weak acid).

(Choice B) Strong acids ionize essentially 100% into H^+ and A^- . If HCN were a strong acid, the $[H^+]$ would be very close to 1×10^{-3} M (ie, the initial HCN concentration).

(Choice C) Although the cyanide ion CN^- can act as a **weak base**, HCN *donates protons* (ie, acts as a weak acid). The calculated value of $[H^+] = 1 \times 10^{-6}$ M (**pH 6**) is *greater* (more acidic) than a neutral solution ($[H^+] = 1 \times 10^{-7}$ M, pH 7) and confirms that HCN acts overall as a weak acid rather than as a weak base.

(Choice D) A strong base would fully dissociate and provide a large $[OH^-]$ and a very small $[H^+]$ with a solution pH much higher than 7. However, the pH cannot equal the pK_a unless $[HCN] = [CN^-]$. In the given 1×10^{-3} M HCN(aq) solution, $[H^+] = [CN^-] = 1 \times 10^{-6}$ M (ie, $[CN^-] < [HCN]$), which shows that HCN(aq) is a *weak acid*.

Educational objective:

In aqueous solutions, strong acids ionize essentially 100%, giving large values of K_a and high $[H^+]$. Weak acids ionize only to a small extent, giving small values of K_a and $[H^+]$ that are much lower than the concentration of the non-ionized acid.

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$$1 \times 10^{-5} = \frac{[\text{OH}^-][\text{HCN}]}{[\text{CN}^-]} = \frac{(x)(x)}{(0.001 - x)} = \frac{x^2}{(0.001 - x)}$$

When x is small, the expression can be simplified by approximation as:

$$1 \times 10^{-5} = \frac{x^2}{(0.001 - x)} \approx \frac{x^2}{0.001} \text{ where } x = [\text{OH}^-] = [\text{HCN}]$$

Solving for x gives the $[\text{OH}^-]$:

$$x^2 = (1 \times 10^{-5})(0.001) = (1 \times 10^{-5})(1 \times 10^{-3}) = 1 \times 10^{-8}$$

$$x = \sqrt{1 \times 10^{-8}} = 1 \times 10^{-4} = [\text{OH}^-]$$

From this $[\text{OH}^-]$ value, the pOH can then be determined:

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(1 \times 10^{-4}) = 4$$

Using pOH then enables the calculation of pH by the relationship between **pOH and pH**:

$$\text{pH} + \text{pOH} = 14$$

$$\text{pH} = (14 - \text{pOH}) = (14 - 4) = 10$$

(Choice A) The first expression should have an x^2 term because $[\text{OH}^-] = [\text{HCN}]$ (ie, they form in a 1:1 mole ratio). A 0.001 M $\text{HCN}(aq)$ solution (ie, a weak acid) has a pH of 6, but this is not the pH of a solution of 0.001 M $\text{KCN}(aq)$ (ie, a salt of the **conjugate base** of HCN). Note that because KCN acts as a base, any pH below 7 may be ruled out without calculations.

(Choice B) The first expression should have an x^2 term (not $2x$) because the K_b multiplies $[\text{OH}^-]$ and $[\text{HCN}]$ rather than adding them. $\text{HCN}(aq)$ has a $\text{p}K_a$ of 9, but this is not the same as the pH of a $\text{KCN}(aq)$ solution.

(Choice C) The first two math expressions are correct; however, the stated pH is incorrect. The **pH** depends on the concentration of $\text{H}_3\text{O}^+(aq)$ (ie, the $\text{H}^+(aq)$ ions) present in solution, not on the $[\text{OH}^-]$ which relates to the pOH.

Educational objective:

A water-soluble salt formed from the conjugate base of a weak acid dissociates fully in an aqueous solution. Because the salt contains the conjugate base of a weak acid, the resulting solution often exhibits a basic pH with a corresponding base dissociation constant K_b .

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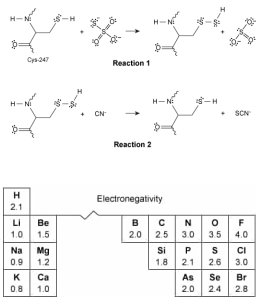
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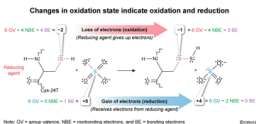


Based on the selected electronegativity values given above and the information in the passage, when Reaction 1 occurs, the atom acting as the reducing agent is the:

- ☐ A. terminal H atom in the S–H bond of Cys-247. [3%]
- ☒ B. substituent S atom in the S–H bond of Cys-247. [46%]
- ☐ C. central S atom in the S=O bonds of S₂O₃²⁻. [32%]
- ☐ D. charged O atom in the S–O⁻ bond of S₂O₃²⁻. [17%]

Correct
46% answered correctly

Explanation:



A **reducing agent** gives up electrons and causes a **reduction** in another compound. In the process, the **reducing agent** itself undergoes **oxidation** by losing the electrons. Oxidation and reduction can be identified in a reaction by examining changes in each atom's **oxidation state**, which is defined as:

oxidation state = (group valence) – (nonbonding electrons) – (assigned bonding electrons)

The **oxidation state** is conceptually **different than the formal charge** of an atom, and the two values are **often not the same number**. **Formal charge** assesses the charge distribution resulting from an atom's bonding configuration by assuming that all bonding electrons are equally shared (divided) between each pair of atoms. In contrast, the oxidation state tracks the gain or loss of electrons relative to an atom's **valence number** by **assigning all bonding electrons** in a bond to the **most electronegative atom**. When the atoms in the bond are the same, they are each assigned one electron per bond.

Conceptually, Reaction 1 depicts a cysteine residue forming a **disulfide bond**, which is an oxidation event. This can be confirmed by **evaluating the oxidation states** of the relevant S atoms. Before Reaction 1 occurs, the oxidation states are:

Substituent S_{Cys-247}(reactant) = (6)_{Group valence} – (4)_{Nonbonding electrons} – ((2)_{Electron assigned from C – S bond} + (2)_{Electrons assigned from S – H bond}) = 6 – 4 – 4 = – 2

Central S_{S₂O₃²⁻}(reactant) = (6)_{Group valence} – (0)_{Nonbonding electrons} – ((1)_{Electron assigned from S – S bond} + (0)_{Electrons assigned from S – O and S = O bonds}) = 6 – 0 – 1 = + 5

After the reaction, the oxidation numbers become:

Substituent S_{Cys-247}(product) = (6)_{Group valence} – (4)_{Nonbonding electrons} – ((2)_{Electron assigned from C – S bond} + (1)_{Electrons assigned from S – S bond}) = 6 – 4 – 3 = – 1

$$\text{Central S}_{\text{(product)}}^{\text{SO}_3^{2-}} = (6)_{\text{Group valence}} - (2)_{\text{Nonbonding electrons}} - \left((0)_{\text{Electrons assigned from S-O bond}} + (0)_{\text{Electrons assigned from S=O bonds}} \right) = 6 - 2 - 0 = +4$$

The bonding electrons in S–S bonds are divided evenly because the electronegativities of the atoms are the same, but the bonding electrons are all taken by the more electronegative atom in C–S, S–H, S–O, and S=O bonds.

The substituent sulfur atom in Cys-247 goes from an oxidation state of –2 to –1 (a loss of an electron) whereas the central sulfur atom in $\text{S}_2\text{O}_3^{2-}$ goes from an oxidation state of +5 to +4 (a gain of an electron). Therefore, the substituent S atom in Cys-247 acts as the **reducing agent** (ie, causes reduction to occur in $\text{S}_2\text{O}_3^{2-}$) but the S atom in Cys-247 itself **gets oxidized in the process**.

The same conclusion can be reached more simply by examining the **bonding partners** of the sulfur atoms. The substituent S atom in Cys-247 loses a bond to H (ie, gets oxidized), and the central S atom in $\text{S}_2\text{O}_3^{2-}$ loses an S–S bond (ie, gets reduced) in the process.

(Choices A and D) The oxidation states of the hydrogen and oxygen atoms do not change in the reaction (ie, these atoms do not participate in oxidation or reduction).

(Choice C) The central S atom in $\text{S}_2\text{O}_3^{2-}$ *takes* electrons and gets *reduced*; therefore, it is causing oxidation in another compound and is acting as an *oxidizing agent*.

Educational objective:

A reducing agent causes a reduction in another compound by giving electrons to the other compound. Therefore, a reducing agent itself undergoes oxidation due to the loss of electrons. The oxidation state accounts for electrons gained or lost by a particular atom.

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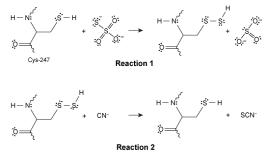
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The experimental LD₅₀ data for rats in Table 1 suggest that HCN may be the most toxic of the three cyanide compounds tested. Does the molar dose data also support this conclusion?

- ☐

A. Yes, because the LD₅₀ and molar dose of KCN are both higher than those of HCN. [12%]
- ☐

B. Yes, because the LD₅₀ of HCN is lower than the LD₅₀ of NaCN. [40%]
- ☐

C. No, because the LD₅₀ and molar dose of NaCN are both lower than those of HCN. [9%]
- ☒

D. No, because NaCN has a higher LD₅₀ but a lower molar dose than HCN. [38%]

Correct

38% answered correctly

Explanation:

Comparing doses to the same species on a molar basis

Animal species	LD ₅₀ (mg/kg)	Molar dose (mmol/kg)	CN ⁻ Compound
Black vulture	4.80	0.098	NaCN
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NaCN is likely the most toxic because it requires the lowest number of millimoles per kilogram to reach LD₅₀.

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An LD₅₀ value is the mass (milligrams) of a substance per kilogram of body weight that is necessary to kill 50% of an animal test group. Lower LD₅₀ values generally indicate greater toxicity, but this can be misleading if the molar masses of the compared substances are very dissimilar. In such cases, a better comparison is made by dividing the LD₅₀ masses by the molar mass to determine the number of moles delivered in the dose.

Comparing the experimental results from Table 1 for the same species shows that the HCN form of cyanide gave the lowest LD₅₀ (by mass) of the three forms of cyanide delivered to rats. This initially seems to indicate that HCN is the most toxic of the three cyanide compounds; however, the molar masses of the compounds are very different. As such, using the molar mass of each compound to convert the LD₅₀ milligrams to millimoles gives a better comparison based on the number of cyanide compound molecules delivered per kilogram.

$$5.72 \frac{\text{mg NaCN}}{\text{kg (rat)}} \times \frac{1 \text{ mmol NaCN}}{49 \text{ mg NaCN}} = 0.117 \frac{\text{mmol NaCN}}{\text{kg (rat)}}$$
$$9.69 \frac{\text{mg KCN}}{\text{kg (rat)}} \times \frac{1 \text{ mmol KCN}}{65 \text{ mg KCN}} = 0.149 \frac{\text{mmol KCN}}{\text{kg (rat)}}$$
$$3.62 \frac{\text{mg HCN}}{\text{kg (rat)}} \times \frac{1 \text{ mmol HCN}}{27 \text{ mg HCN}} = 0.134 \frac{\text{mmol HCN}}{\text{kg (rat)}}$$

The molar doses show only slight differences in the toxicity of the compounds, but the results suggest that fewer millimoles of NaCN are required for 50% mortality. Therefore, if these differences are statistically significant, NaCN is likely the most toxic of the three compounds.

(Choice A) The LD_{50} and molar dose of KCN are indeed both higher than those of HCN; however, the molar dose of NaCN is lower than that of HCN, which may indicate that HCN is not the most toxic of the three compounds.

(Choice B) Although HCN has the lowest LD_{50} , a comparison of the doses on a molar basis shows that fewer molecules (ie, a lower molar dose) of NaCN is required to result in 50% mortality.

(Choice C) LD_{50} of NaCN is *higher* than the LD_{50} of HCN, which supports the conclusion. Only the molar dose of NaCN is lower than that of HCN and contradicts the conclusion.

Educational objective:

An LD_{50} value gives the mass (milligrams) of a substance per kilogram of body weight necessary to kill 50% of an animal test group. Lower LD_{50} values generally indicate a more toxic substance, but this comparison can be misleading if the molar masses of the compared substances are very dissimilar. A better comparison may be achieved by an evaluation of the doses on a molar basis.

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